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* TCP/IP Model

- The TCP/IP (Transmission Control Protocol/ Internet Protocol) is the fundamental communication architecture of internet. It defines how data is packaged, addressed, transmitted, routed & received across networks.

- It is protocol-based, meaning that it specifies a set of rules (protocols) that computers & peripherals like routers, switches, network cards, printers, etc. use to communicate reliably.

- It is practical with 4 layers & is the one actually used in real-world meaning.

→ Layers of TCP/IP Model:-

1) Application Layer: Closest to the user. Provides network services to applications. Defines how software applications communicate over a network. Handles high-level protocols for file transfer, email, browsing, etc. Protocols used are HTTP/HTTPS, FTP, SMTP, DNS.

2) Transport Layer: Provides end-to-end communication between devices (host-to-host). Splits application data into segments. Ensures reliability, sequencing & error-checking. TCP (Transmission Control Protocol) & UDP (User Datagram Protocol). Peripherals involved are operating system kernel (TCP/UDP stacks), NIC (Network Interface Cards)

3) Internet Layer: Responsible for logical addressing & routing. It assigns IP addresses & determines best path from source to destination across multiple networks. IP (Internet Protocol) used for addressing & routing. Peripherals involved are routers, Layer 3 switches, firewalls

4) Network Access Layer (or Link Layer): Responsible for physical transmission of data over the network. Converts IP packets into frames & transmits over physical media (cables, wifi, fiber). Handles MAC addresses, error detection at hardware level. Peripherals involved are Network Interface cards (NICs), hubs, switches, modems, access points, physical cables.

* Network & its types

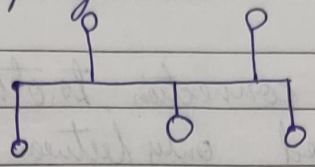
- A network is a collection of two or more interconnected devices (servers, computers, printers, smartphones or peripherals) that communicate & share resources like data, files, internet, applications using transmission media & communication protocols.

→ The internet is a type of world-wide computer network. The Internet is a connection of infinite numbers of connected computers that are spread across the world.

→ Network topology is the arrangement of the various components like links, nodes, etc of a computer network.

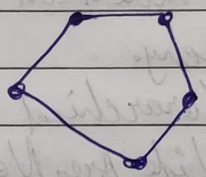
* Types of network topologies :

1) Bus Topology



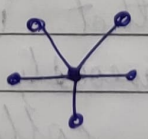
- Every computer & network device is connected to a single cable.
- It transmits data only in one direction.
- Cost effective
- Used in small networks.
- Easy to expand joining two cables together
- It is used in early LAN connection.

2) Ring Topology



- It forms a ring as each computer is connected to another computer, with the last one connected to the first.
- Transmission is unidirectional & sequential way that is bit by bit.
- Transmitting network is not affected by high traffic or by adding more nodes, as only the nodes having tokens can transmit data.
- Cheap to install & expand.

3) Star Topology



- Computers are connected to a single central hub through a cable.
- Fast performance with few nodes & low network traffic.
- Easy to troubleshoot & easy to setup & modify.
- Hub can be upgraded easily.

4) Mesh Topology

- Point-to-point connection to other devices or fully connected.
- Traffic is carried only between two connected devices.
- Robust, costly but not flexible.
- Fault is diagnosed easily.
- More cable resource used in setup.

5) Tree Topology

- It has a root node & all other nodes are connected to it forming a hierarchy.
- Also called Hierarchical topology
- Mostly used in Wide Area Network - WAN.
- Expansion of nodes is possible & easy.
- Easily managed & maintained.

* CSMA/CD (Carrier Sensor Multiple Access with Collision Detection)

- It is a network protocol used in Ethernet to manage how devices share the same communication channel. It ensures that when multiple devices try to send data simultaneously on a shared medium, the system can detect collisions and resolve them.

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Q1)

* **Uses of Computer Networks:-**

- Hardware devices like printers, scanners & external storage can be shared across multiple computers.
- Instead of installing software on every computer, centralized applications can be run from servers.
- Users can easily share files, documents & media across the network.
- Networks enable instant communication through email, chat apps & video conferencing.
- Services like Skype, WhatsApp & Zoom rely on networks to transmit voice & video over data using Voice over IP
- Teams can work on shared documents & projects in real time like Google Docs.
- Large organizations store & manage data in central servers for easy access & security.
- Cloud computing services like AWS, Google Cloud & Microsoft Azure allow businesses & individuals to use powerful computing resources without owning the ~~to~~ hardware.

Q2) Define computer network & list 3 common applications

- A computer network is a system in which two or more computer networks & peripheral devices are connected together using communication channels (wired or wireless) to share resources, exchange data and enable communication.

Three common applications are :-

- Users can share files, printers & storage devices across the network.
- Email, messaging & video conferencing are enabled through computer networks.
- Networks provide access to the internet, cloud computing, e-commerce & multimedia streaming.

Q3) Uses of bridges in computer network.

Ans3) A bridge is a networking device that connects & filters traffic between two or more LAN segments. It works at the Data Link Layer (Layer 2) of the OSI model. Its main role is to forward useful data from one network segment to another while reducing unnecessary traffic.

- Bridges divide a large network into smaller segments which reduces traffic congestion by limiting unnecessary data broadcasts.

- Traffic Filtering: Bridges analyze the MAC addresses of data packets. They only forward data to segment where the destination device exists. This improves efficiency & reduce collisions.

- Extending Networks: Bridges can connect two physically separate LANs, making them act as one large network. Useful when expanding office spaces or linking different floors of a building.

- By reducing collision domains, bridges improve overall network speed & performance.

- Bridges are usually protocol-independent. They simply forward data based on MAC addresses, regardless of higher-level protocols (like TCP/IP).

24 September 2025 – Wednesday

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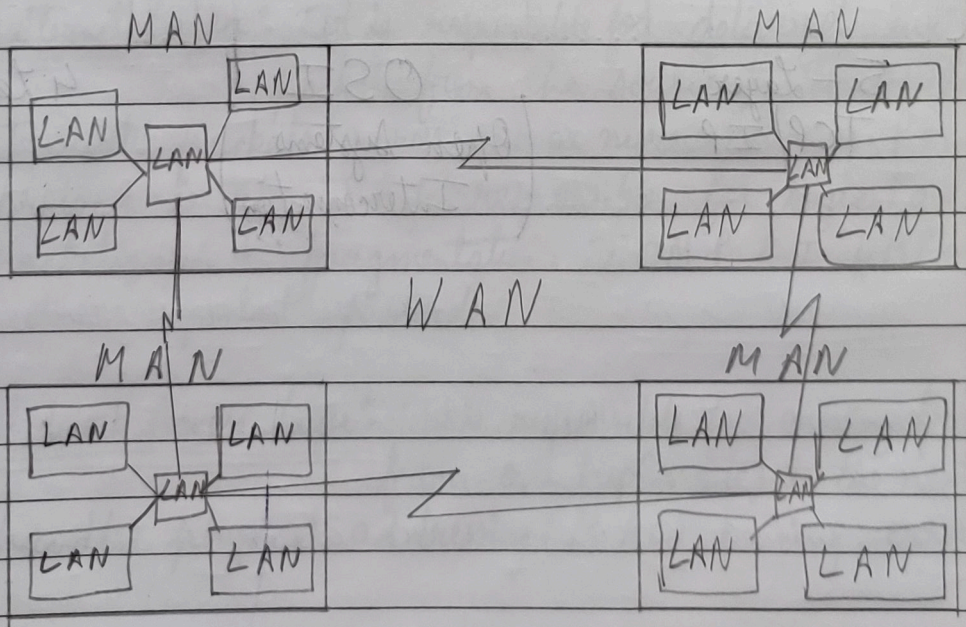
- A computer network is a system in which multiple computers are connected to each other to share information & resources.
- Connections can be wired (cables, switches, routers) or wireless (wifi, Bluetooth). It also includes network devices that help in communication.

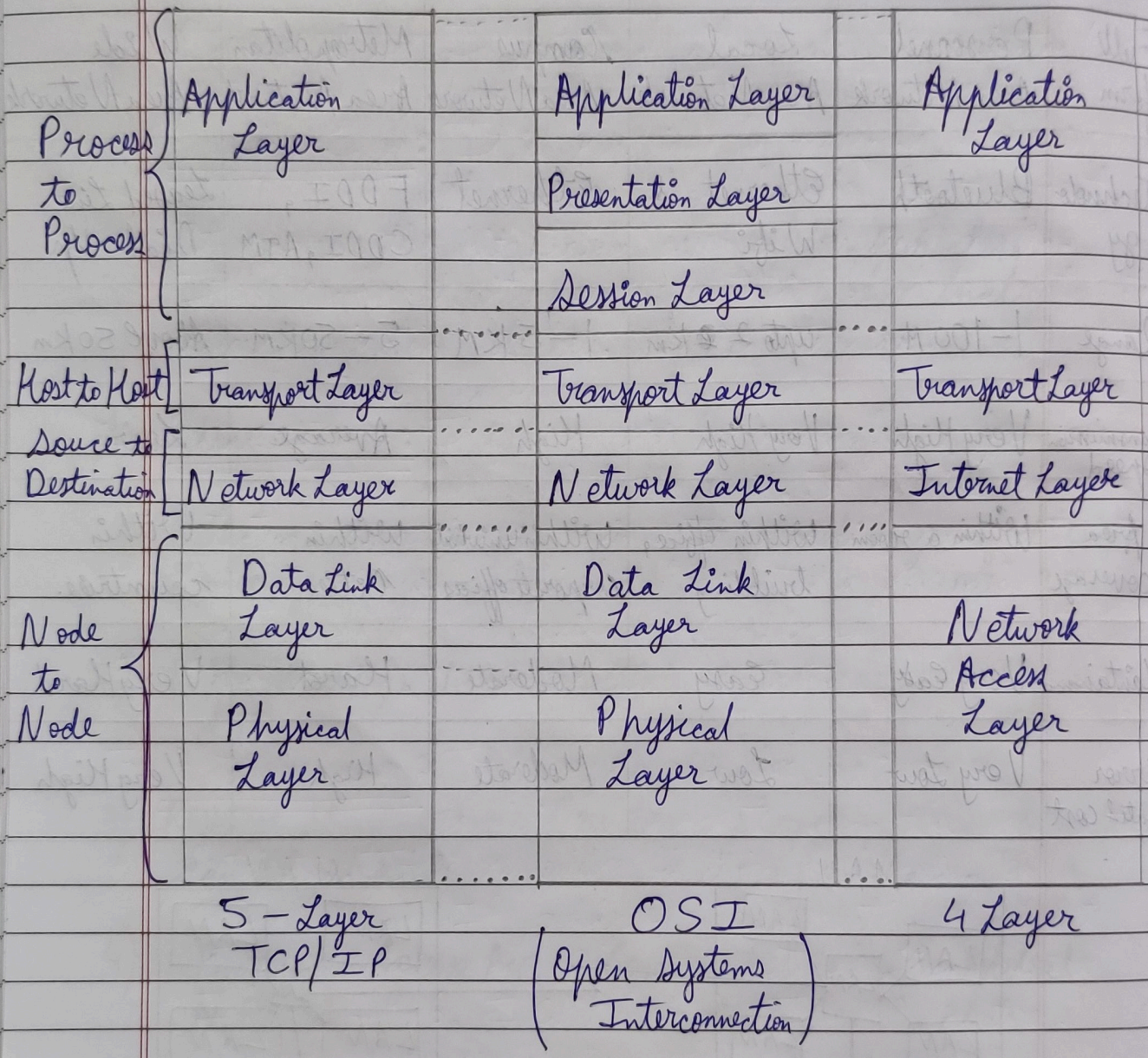
✳ Uses

1) Better communication i.e. easy exchange of data/ message.
2) File sharing i.e. multiple users can share files & data.
3) Resource sharing i.e. Printers, storage devices & internet can be shared.
4) Internet Access i.e. Connection to the Global Network
5) Flexible Access i.e. access resources from anywhere.
6) Entertainment i.e. streaming, gaming, multimedia
7) It reduces cost of hardware / software by sharing.

	PAN	LAN	CAN	MAN	WAN
Full form	Personnel Area Network	Local Area Network	Campus Area Network	Metropolitan Area Network	Wide Area Network
Technology	Bluetooth	Ethernet & Wifi	Ethernet	FDDI, CDDI, ATM	Leased Line Dial-Up
Range	1-100 M	upto 2 Km	1-5 KM	5-50KM	Above 50km
Transmission Speed	Very High	Very High	High	Average	Low
Area coverage	Within a room	Within office, building	Within university corporate offices	Within Cities	Within countries.
Maintainance	Very Easy	Easy	Moderate	Hard	Very Hard
Error Rate & cost	Very Low	Low	Moderate	High	Very High



* TCP/IP Protocol Suite or Internet Protocol



* 4 Layer Model

1) **Application Layer**: It provides network services directly to user applications (web browsers). This is allowing access to network resource. It provides services like network virtual terminal, file transfer, access & managment, mail services, etc.

2) **Transport Layer**: It is responsible for the delivery of message from one process to another. This layer ensures that the whole message arrives intact & in order. It provides servicds like multiplexing/demultiplexing, reliability, flow control & segmentation/reassembly. Two core protocols are:-

   1) TCP – Transmission Control Protocol

   2) UDP – User Datagram Protocol.

3) **Internet Layer**: It is responsible for delivery of individual packets from the source to ~~host~~ destination. It routes packets across one or more networks. Primary protocol is IP, IP is responsible for logical addressing, packetization & fragmentation. ICMP & IGMP are other important protocols.

4) **Network Access Layer**: It is responsible for moving frames from one hop(node) to the next. It handles physical addressing (MAC), framing, error detection.

* OSI Model (7 layers)

- Developed by International Standards Organization (ISO)

1) **Physical Layer:** It is responsible for movements of individual bits from one hop (node) to the next. It carries the bit stream over a physical media. It is concerned with representation of bits, data rate, synchronization of bits, physical topology.

2) **Data Link Layer:** It is responsible for moving frames from one hop (node) to the next. It provides error detection, correction & framing. It organizes raw bits into frames carried from physical layer. It is responsible for framing, physical addressing (MAC addresses), flow control, error control, and access control.

3) **Network Layer:** It is responsible for delivery of individual packets from the source to destination. It is responsible for logical addressing eg 192.68.1.1 (IP address) and routing of packets between devices on different networks. Examples are IPv4, IPv6, ICMP (ping), IGMP, routers.

4) **Transport Layer:** It delivers the message from one process to another. It provides end-to-end communication between applications. It is responsible for segmentation & reassembly of data. It ensures that the whole message arrives intact & in order. Examples are TCP (Transmission Control Protocol) & UDP (User Datagram Protocol).

5) Session Layer: It is responsible for dialog control & synchronization. This layer acts as a network dialog controller — establishes, maintains, synchronizes the interaction among computers. It is the manager of sessions between two systems.

6) Presentation Layer: It is responsible for data translation, compression, encryption. This layer is concerned with the syntax which refers to order in which data is presented & semantics helps in interpreting a particular pattern.

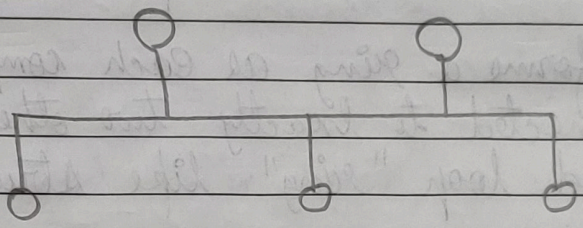
7) Application Layer: It is responsible to provide network services to the user. It provides services like file transfer, access & management, mail services, directory services. It defines high level protocols like HTTP/HTTPS, FTP, SMTP, DNS

	OSI Model	TCP/IP Model
1)	It has 7 layers	It has 5 layers
2)	It provides layer functioning. Each layer has specific, well defined ~~fun~~ functions.	It is based on protocols & protocols are not flexible with other layers.
3)	The Transport layer gaurantees delivery of packets.	The Transport Layer does not gaurantee delivery of packets.
4)	It follows horizontal way.	It follows vertical way.
5)	More prescriptive: each layer has a narrowly defined role.	Less rigid — some functions are spanning layers.
6)	It provides both connection oriented & connectionless service in	It provides connectionless service.
7)	Protocols are hidden OSI layers & are easily replaced	Protocols are explicit & cannot be replaced easily
8)	It defines services, interfaces & protocols very clearly & makes clear distinction between them.	It does not clearly seperates its services, interfaces & protocols.

* Topologies

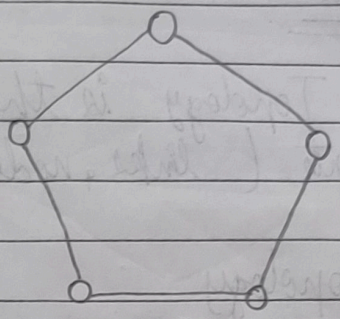
Network Topology is the arrangement of the various components (links, nodes, etc) of a computer network.

1) Bus Topology



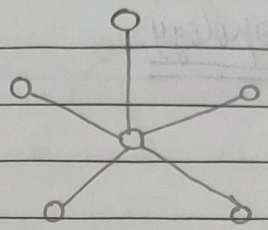
- Every computer & network device is connected to a single cable.
- It transmits data only in one direction.
- Cost effective
- Used in small networks
- Easy to expand, by joining two cables together.
- It is used in early LAN connection.

2) Ring Topology



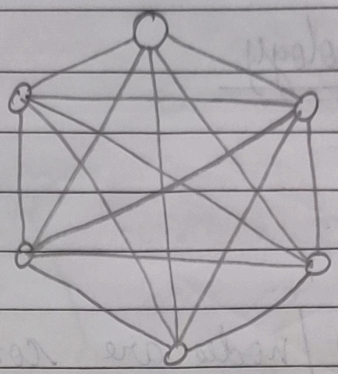
- It forms a ring as each computer/device is connected to exactly two other devices, forming a closed loop "ring" like structure.
- Data travels in one direction or both directions, passing through each device until it reaches its destination.
- Simple & cheap easy to install & expand
- Data moves in circular path.
- Performs better than bus topology under heavy network load.
- No central hub/switch is required unlike star topology.
- Failure of a single node/cable can break the entire network.
- Difficult to troubleshoot & reconfigure.
- Slower than star topology in large networks.

3) Star Topology



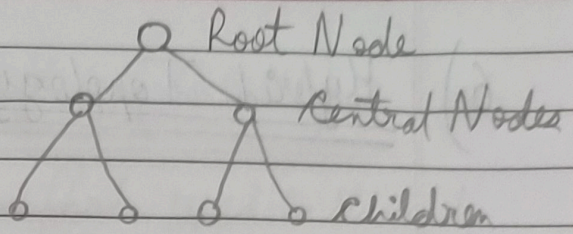
- All devices / nodes are connected to a central device through a cable.
- Communication between two nodes passes through the central device (hub / switch / router).
- Each node has a point - to - point connection to the central device.
- Failure of one node does not affect the rest of network.
- Centralized managment makes troubleshooting easy.
- Provides better performance than bus or ring since nodes dont share a single cable.
- If the central hub / switch fails, the whole network stops.
- Requires more cable than bus or ring topology.
- Cost is higher due to central networking devices.
- Example: Most modern ethernet LANs use star topology with switches. Common in office networks.

4) Mesh Topology



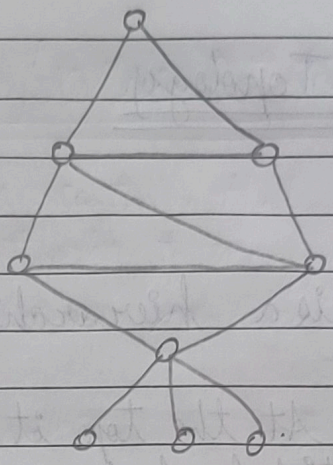
- Every device is connected directly to every other device in the network.
- It has multiple paths for data travel, ensuring high redundancy & reliability.
- It provides maximum redundancy but is expensive.
- Connection is carried only between two connected devices.
- Fault is diagnosed easily.
- More cable resource used in setup.
- Complex setup & maintenance; harder to manage.
- Not scalable for very large networks.
- Robust & fast communication.
- Highly reliable: even if one link fails, data can take another path.

5) Tree Topology



- It is a hierarchial network topology.
- ~~It~~ At the top it has root node, below it are multiple levels of secondary node connected in a ~~star~~ tree like manner.
- It has root node connected to one or more central nodes.
- Central nodes act as parents & devices connected to them act as children.
- Expansion happens level by level, like branches of a tree.
- Easy to add more nodes by extending branches.
- Each level can be managed separately.
- If one branch fails, only that branch is affected.
- If Root node fails, entire network fails.
- Complex installations & maintenance as the network grows.
- It is a combination of star & bus.

6) Hybrid Topology



- It is combination of 2 or more different network topologies like star, bus, mesh, tree, ring to form a single, efficient network.
- It is designed to take the advantages of each topology & minimize their disadvantages, making it the most optimized network for large & complex ones.
- It provides flexibility & scalability.
- Design depends on the organization's needs.
- Highly reliable; a failure in one part doesn't necessarily break the whole network.
- Complex design & installation
- High cost (cabling + device)
- Maintenance is difficult due to multiple types of connections.

* Carrier Sense Multiple Access (CSMA)

It is a network access method used in shared communication channels like Ethernet or Wifi where multiple devices share the same medium.

Before sending data, a device "listens" (carrier sense) to check if the channel is free.

- If the medium is idle, it transmits.
- If the medium is busy, it waits until it becomes free.

This helps reduce collisions when multiple devices try to send data at the same time.

→ Carrier Sense: Each device listens to the medium before transmitting.

→ Multiple Access: Many devices share the same communication channel.

→ Collision Possibility: Even with listening, two devices may still transmit almost simultaneously in which ~~result~~ collision may occur.

- It reduces collisions compared to pure random access like ALOHA.
- It is simple & decentralized. No ~~Extend~~ central controller needed.

* Carrier Sense Multiple Access / Collision Detection (CSMA/CD)

CSMA/CD is a media access control method used mainly in wired ethernet networks to manage how multiple devices share the same communication channel.

It combines two ideas:

1) CSMA (Carrier Sense Multiple Access): each device listens to the channel before transmitting.

2) CD (Collision Detection): If two devices still transmit at the same time and a collision occurs, they detect it, stop and retransmit later.

* Working

- Device checks if the channel is idle.
- If free, it starts transmitting.
- While transmitting, it listens for changes to detect if another device is also transmitting.
- If collision detected, devices send a jam signal to notify others.
- Each device waits for a random backoff time
- After waiting, devices sense the medium again and retransmit.

* It prevents data corruption by detecting collisions early.

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* Carrier Sense Multiple Access with Collision Avoidance

CSMA/CA is a media access contrd method mainly used in wireless networks (wifi) to reduce the chance of collisions when multiple device share the same channel.

Unlike CSMA/CD, wireless devices cannot detect collisions while transmitting. So instead of detecting, CSMA/CD tries to avoid collisions before they happen.

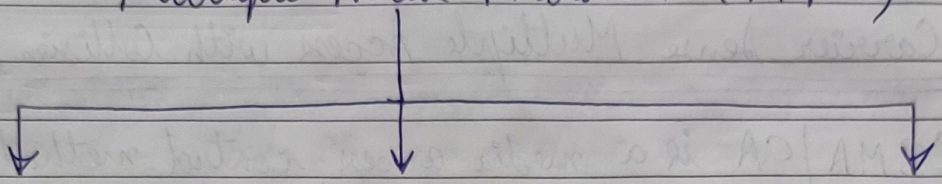
* Working

- Device listens to the channel to check if its free.
- If idle, device waits for a short period, then sends data.
- If busy, device waits for a random backoff time before checking again.
- Reciever sends an ACK frame if data is recieved correctly, ensuring reliability.

* Advantages

- It reduces chances of collisions in wireless networks.
- Works well with many devices in the same wifi area.
- Supports reliability with ACK mechanism.

Multiple Access Protocols (MAC)



Random Access Protocols	Control Access	Channelization Protocols
• Aloha	• Polling	• FDMA
• CSMA	• Token passing	• TDMA
• CSMA/CA		• CDMA
• CSMA/CD		

* Pure Aloha

- It is a simple, random access protocol used in computer networks where multiple devices share a common communication channel.
- In pure ALOHA, a device can transmit data at any time without checking if channel is free.

* Working

1) A device sends a frame whenever it has data.
2) After sending, it waits of an acknowledgement (ACK) from the reciever.
3) If ACK is received i.e. transmission successful
4) If no ACK i.e. collision occured & device waits

a random backoff time & retransmits.

☆ Since devices don't sense the medium before sending, collisions are frequent.

- The throughput of the Pure ALOHA is maximized when the frames are of uniform length.
- The formula to calculate throughput of the Pure ALOHA is

$$S = G * e^{-2G}$$

Here, G = number of stations transmitting data
e = constant

Throughput is maximum when G = 1/2 which is 18% of the total transmitted data frames.

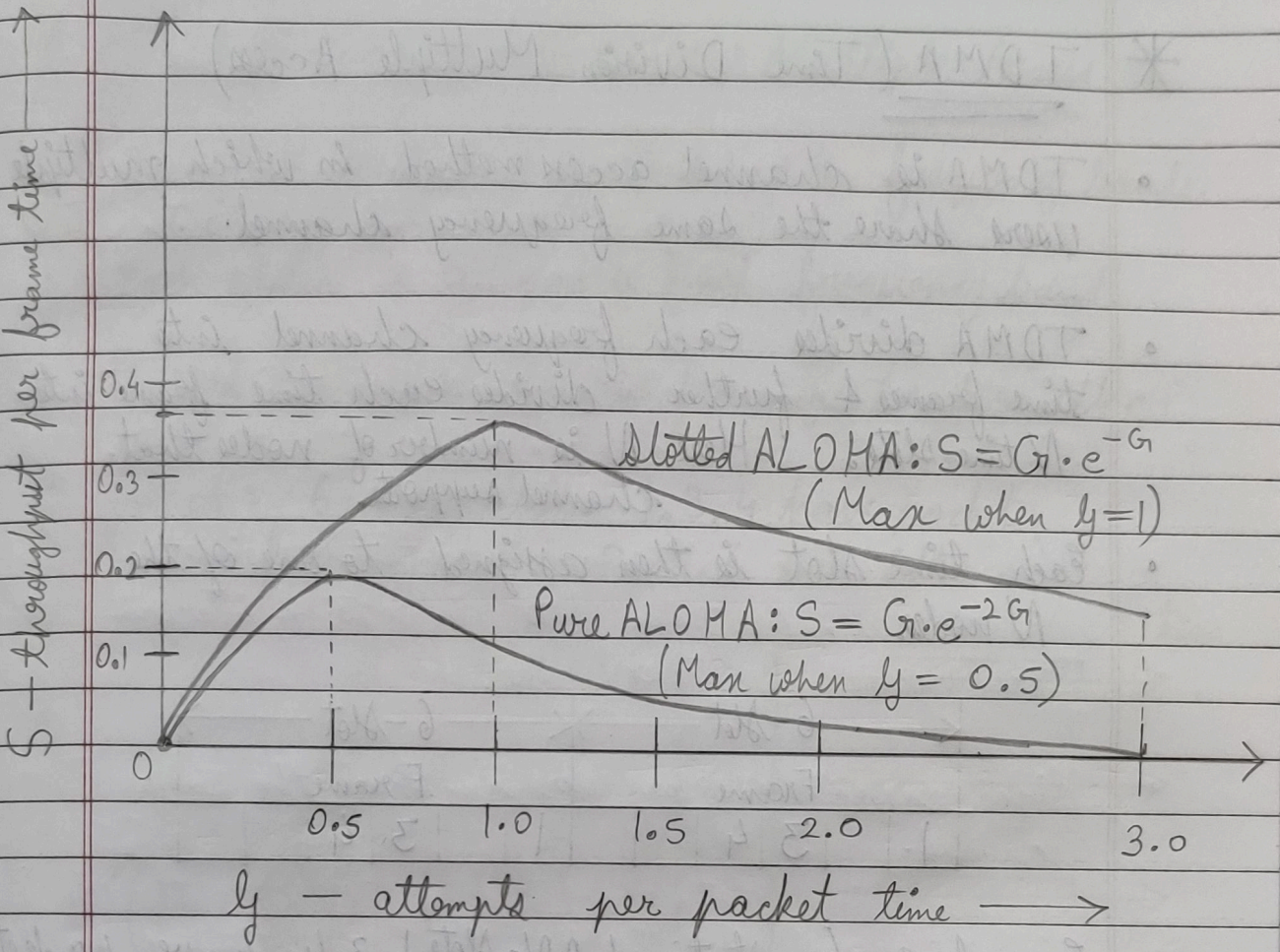
- Works well with low traffic networks
- Useful in satellites & wireless systems with long propagation delays.
- High collisions probability under heavy load.
- ~~No~~ Not efficient for large scale networks.

* Slotted ALOHA

- It was invented to improve the efficiency of pure ALOHA as chances of collisions in pure ALOHA are very high.
- The time of the shared channel is divided into discrete intervals called slots.
- The stations can send a frame only at the beginning of the slot & only one frame is sent in each slot.
- If any station is not able to place the frame onto the channel at the beginning of the slot then the station has to wait until the beginning of the next time slot.
- The formula to calculate the throughput of the Slotted ALOHA is

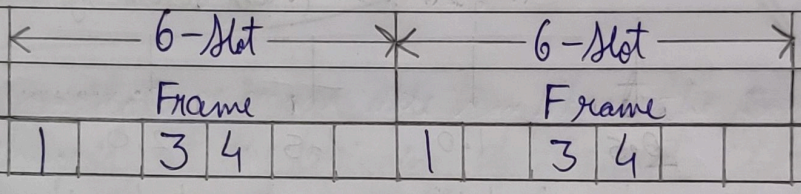
$$S = G * e^{-G}$$

- The throughput is maximum when $G = 1$ which is 37% of total transmitted data frames.
- 37% of the time slot is empty
- 37% successes
- 26% collisions


S – throughput per frame time
0.4
0.3
0.2
0.1
0
Slotted ALOHA: S = G · e^-G
(Max when G=1)
Pure ALOHA: S = G · e^-2G
(Max when G = 0.5)
0.5
1.0
1.5
2.0
3.0
G – attempts per packet time

* TDMA (Time Division Multiple Access)

- TDMA is channel access method in which multiple users share the same frequency channel.
- TDMA divides each frequency channel into time frames & further divides each time frame into N time slots. Here N is number of nodes that channel supports.
- Each time slot is then assigned to one of the N nodes.



- Example: 6-station LAN. Slots 1, 3, 4 have packet slots 2, 5, 6 are idle.

* Advantages

- No collision since slots are pre-allocated.
- Efficient use of bandwidth in high-load systems.
- Simple scheduling for multiple users.

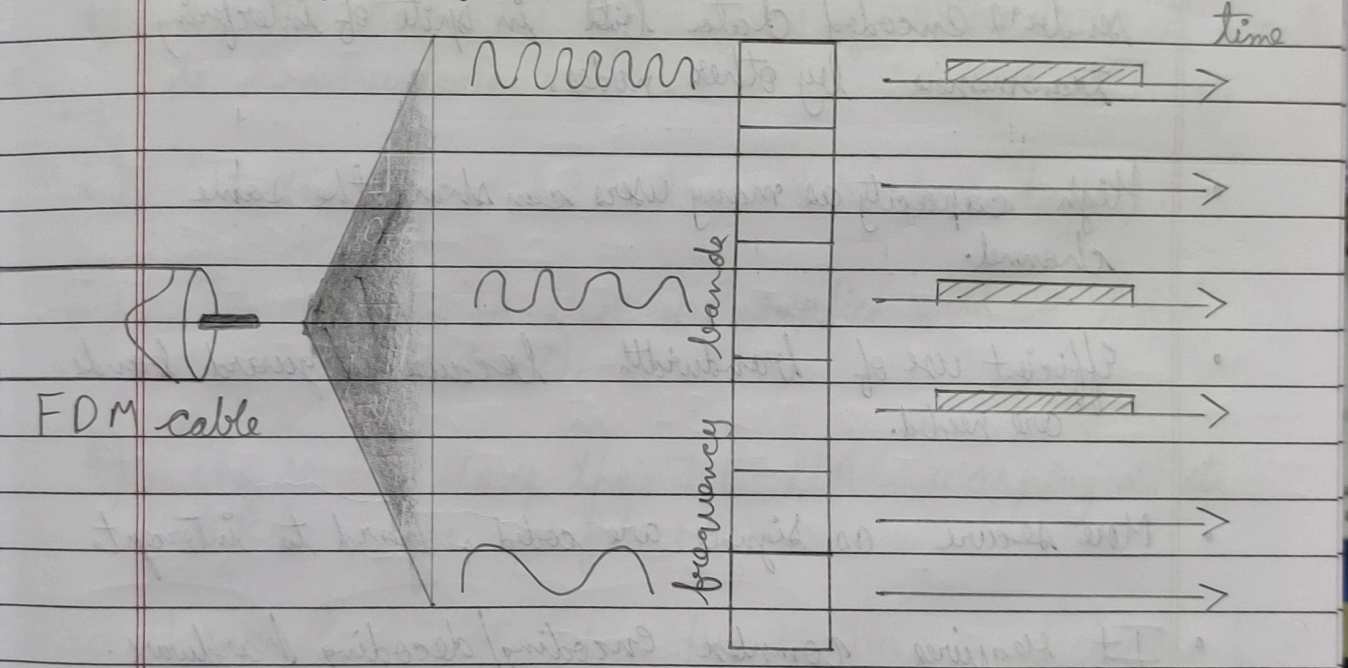
* Disadvantages

- Requires time synchronization between devices.
- If a user has no data to send, their slot is wasted.
- Higher complexity compared to ALOHA.

27 September 2025 – Saturday

* FDMA – Frequency Division Multiple Access

- Channel spectrum is divided into frequency bands.
- Each station is assigned a fixed frequency band.
- Unused transmission time in frequency bands go idle.
- Example: 6-station LAN. 1, 3, 4 have packet, frequency bands 2, 5, 6 are idle.



- All users can transmit simultaneously, but on different frequency channel, so no interference occurs.
- No collisions as each user has its own channel.
- Works well for continuous, steady traffic (like voice)
- Examples are Radio & TV broadcasting, 1G analog mobile systems, Satellite communication.

* CDMA — Code Division Multiple Access

- It assigns a unique code to each node, each node then uses its unique code to encode the data bits it sends.
- If the codes are chosen carefully, CDMA networks can transmit simultaneously from different nodes.
- Their respective receivers correctly receive a sender's encoded data bits in spite of interfering transmissions by other nodes.
- High capacity as many users can share the same channel.
- Efficient use of bandwidth because no guard bands are needed.
- More secure as signals are coded, hard to intercept.
- It requires complex encoding/decoding hardware.

* ARP — Address Resolution Protocol

ARP is a network protocol used to find the MAC address i.e. physical address of a device from its IP address in a Local Area Network (LAN).

ARP maps an IP address to a MAC address.

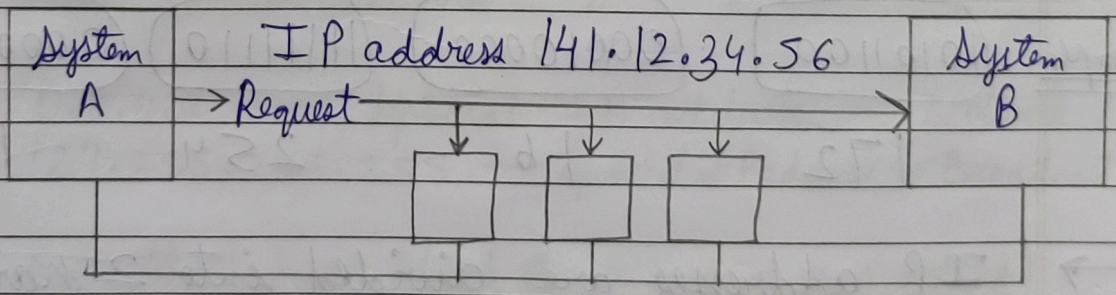
* Working

1) A device knows the IP address of another device it wants to communicate with.

2) It sends an ARP Request (broadcast).

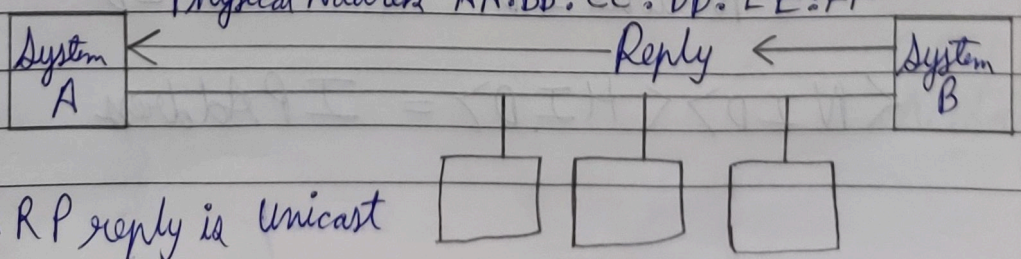
3) The device with that IP replies with an ARP Reply (unicast).

4) The sender stores the IP-MAC mapping in its ARP cache for future use.



ARP request is broadcast.

Physical Address AA:BB:CC:DD:EE:FF



ARP reply is unicast

* IP Address (Internet Protocol Address)

An IP address is a unique numerical identifier assigned to every device connected to a network like the internet or a LAN.

It helps in identifying a device & locating it on the network.

* Functions

- Identification of a device on a network.
- Shows where the device is located in the network.
- Routing the data packets find the correct destination.

→ IPv4: IP addresses are 32 bit numbers which are divided into 4 octets. Each octet represents 8 bit binary number.

Example:

8 bit	8 bit	8 bit	8 bit
10101100	00010000	11111110	00000001
172	16	254	1

→ IP addresses are divided into 2 parts:

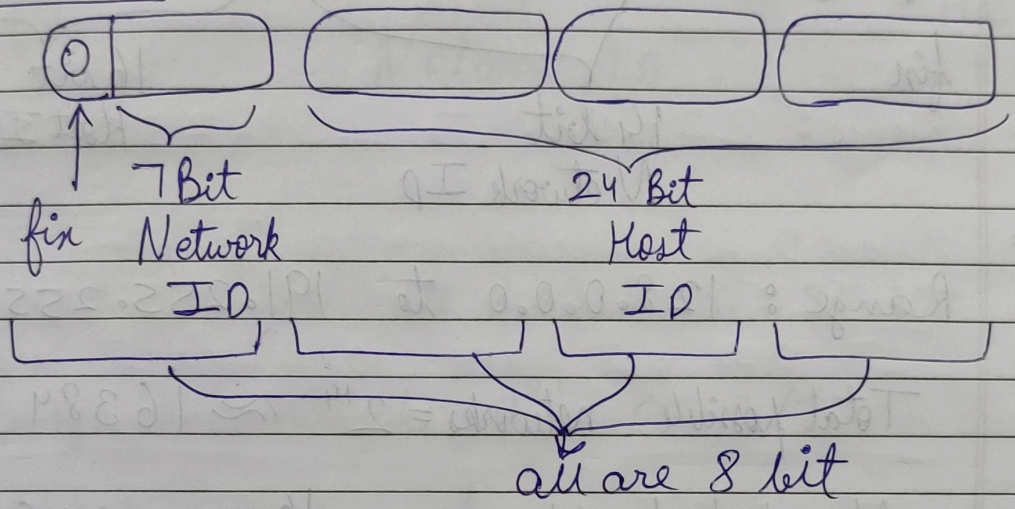
Network ID & Host ID

<NID><HID> = IP Address

* Classes of IPv4 addresses

IPv4 addresses are divided into 5 classes A to E
(32 bit — Dotted decimal format)

1) Class A

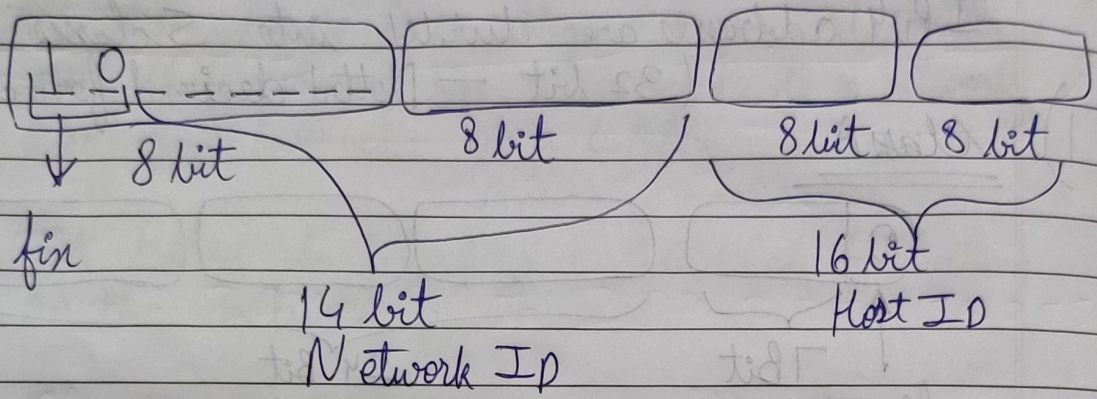


- Range : 1**0**.0.0.0 to 126. 255.255.255
(Useful)
- Only 126 addresses are used for network address.
- Host IPs per network = $2^{24} - 2$
- 0.0.0.0 — Special IP address — Network Number
127.255.255.255 — Special IP address — Broadcast IP.

☆ Default Subnet Mask = 255. 0.0.0 (/8)

CIDR Notation

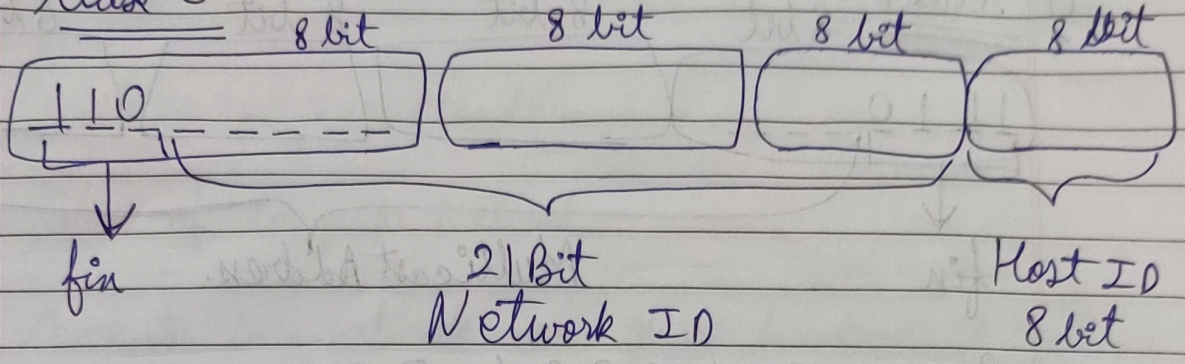
2) Class B



- Range : 128.0.0.0 to 191.255.255.255
- Total possible networks = $2^{14} \approx 16384$
- No of host addresses = $2^{16} - 2 \approx 65534$

☆ Default Subnet Mask = 255.255.0.0 (/16)

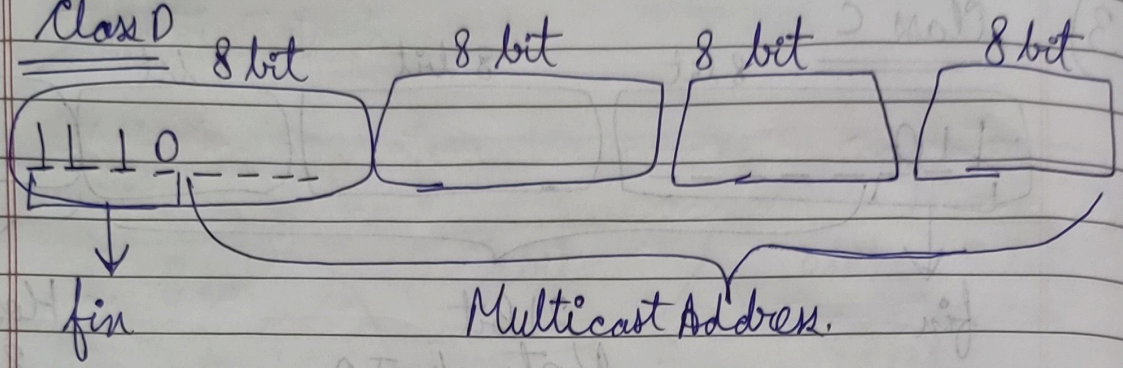
3) Class C



- Range: 192.0.0.0 to 223.255.255.255
- No. of total networks = $2^{21} \approx 2M$
- Hosts per network = $2^8 - 2 = 256 - 2 = \boxed{254}$

☆ Default Subnet Mask = 255.255.255.0 (/24)

4) Class D



- Range: 224.0.0.0 to 239.255.255.255
- Class D is reserved for multicasting
- It does not have any subnet mask.

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6 October 2025 — Monday

* Uses of Computer Network

- Computer in a network can share hardware & software resources, reducing cost & improving efficiency.
- Users can easily share & access data, files & information across the network
- Ensures electronic communication through email, chat, video conferencing

* A computer network is a system of computers interconnected to each other through wires or wireless to share information & resources, exchange data.

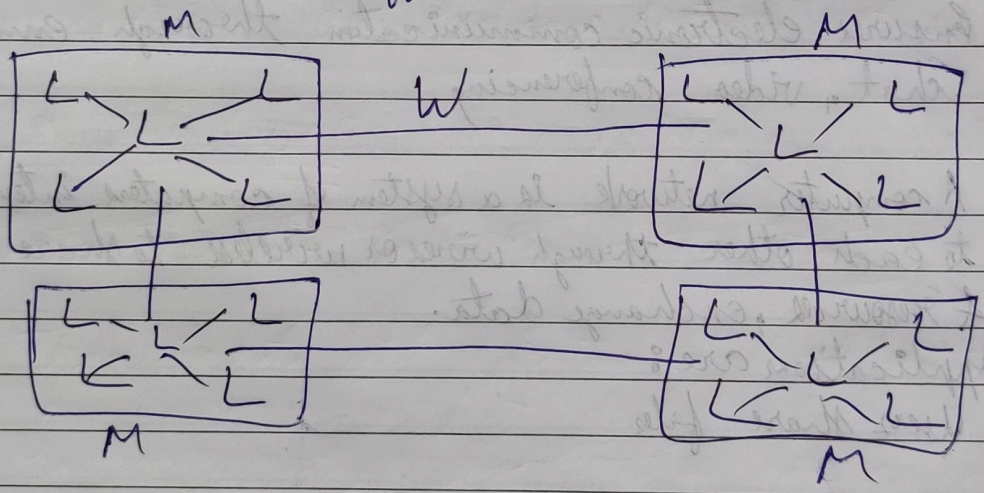
Applications are :-

- Users share files
- Email, message, video conferen
- E-commerce, multimedia stream
- Flexible access from anywhere
- Cloud computing

* Types of Network

- LAN (Local Area Network) interconnects computers within a limited area such as a school or college
- MAN (Metropolitan) interconnects in metropolitan area like city.
- WAN (Wide) it exists over a large-scale geographical area.

WAN connects different LANs & MANs



* **OSI Model** — Open Systems Interconnection.

7	Application
6	Presentation
5	Session
4	Transport
3	Network
2	Data Link
1	Physical

1) **Physical Layer:**

- Transmits raw bits over a physical medium
- It is responsible for representation of bits, data rate, synchronization of bits.

2) **Data Link Layer:**

- Responsible for ~~the~~ moving frames from one node to next.
- Handles flow control, error control, access control.

3) **Network Layer:**

- Determines best path for data routing.
- Provides logical addressing — IP addresses.
- Responsible for delivery of packets.

4) Transport Layer:

- Delivers message from one process to another.
- Ensures reliable data transfer.
- TCP & UDP are examples

5) Session Layer:

- Establishes, manages & terminates sessions.
- Synchronizes data exchange between systems.

6) Presentation Layer:

- ~~Data translation, compression & encryption.~~
- Translates data bet" applications & network formats.
- Handles encryption, decryption & data compression.

7) Application Layer:

- Provides user interfaces & network services
- Supports protocols like HTTP, FTP, SMTP.

* TCP/IP

4 Application Layer
3 Transport Layer
2 Internet Layer
1 Network Access Layer

1) Network Access Layer

- Deals with physical transmission of data
- Defines how data is sent over physical networks
- Handles framing, error detection & physical addressing (MAC)

2) Internet Layer

- Responsible for logical addressing & routing.
- Determine best path for data packets.
- Protocols used are Internet Protocol, ICMP, & ARP.

3) Transport Layer

- Provides end-to-end communication bet^n devices.
- Handles segmentation, flow control & error correction.
- Ensures ~~error~~ reliable data delivery & fast, connectionless service.

4) Application Layer

- Provides services directly to user/applications
- Protocols: HTTP, FTP, DNS, SMTP
- Enables communication bet" software applications over the network.

* MAC address — Media Access Control

- It is a unique hardware identifier assigned to every network interface card (NIC) by the manufacturer.
- It uniquely identifies a device on a local network.
- Used in data transmission within same local area network (LAN).
- MAC Address operates at Data Link Layer (Layer 2) of the OSI Model

Carrier Sense Multiple Access

Collision Detection CSMA/CD	Collision Avoidance CSMA/CA
• Detects collision after it happens & then takes action to retransmit.	• Tries to avoid collision before it happens by using control signals.
• Used in wired network like Ethernet	• Used in wireless networks like Wifi.
• Collisions are detected & retransmission is done after a random delay.	• Collisions are avoided using Acknowledgement (ACK) & back-off mechanisms.
• Works well with many devices within same ethernet cable.	• Works well with many devices within same wifi area.

✳ Slotted ALOHA vs Pure Aloha

→ Pure Aloha

- In pure Aloha, a station can transmit data at any time without checking the channel.
- If 2 or more stations transmit at the same time, collision occurs & data must be retransmitted.
- Formula of throughput

$$S = G * e^{-2G}$$

G = no. of stations
e = constant
S is max @ G = 1/2 i.e. S = 18%

→ Slotted Aloha

- Time is divided into equal slots & a station can only transmit at the beginning of a time slot.
- This synchronization reduces the chances of collisions
- Formula of throughput

$$S = G * e^{-G}$$

S is max @ G = 1 i.e. S = 36.8%

* ALOHA	CSMA/CD
• Stations transmit data whenever ready, without sensing channel	Stations sense channel before transmitting & detect collisions during transmission.
• Collisions are detected after they occur.	Collisions are detected during transmission.
• Low efficiency 18% — Pure ALOHA 36.8% — Slotted ALOHA	Higher efficiency 70 – 80% under normal load.
• Used in wireless or satellite communication	Used in wired communication.
• Difficult to detect collisions in wireless	Easy to detect collisions in wired networks.

* Types of addresses in IPv4 Network

- Network address by which we refer to the network

  eg: 10.0.0.1

- Broadcast address – a special address used to send data to all hosts in the network.

  eg: 10.0.0.255

- Host address – assigned to end devices in the network

  eg: 10.0.0.1

* ARP – Address Resolution Protocol

- Used to find MAC address from the IP in a LAN
- ARP maps IP to MAC
- When a device wants to send data to an IP address on same network, it broadcasts an ARP request asking, "Who has this IP?"
- Device with matching IP replies with its MAC address
- Sender uses this MAC address to deliver data.

*	TCP – Transmission Control	User datagram UDP
•	Connection oriented	Connectionless
•	Reliable, ensures data delivery	Unreliable, no guarantee
•	Uses segmentation, sequencing & acknowledgement to manage data.	Sends data as datagrams without sequencing or acknowledgement.
•	Slower due to error checking	Faster because it has minimal overhead.
•	Suitable for apps requiring ↑ accuracy.	Suitable for apps requiring speed over reliability.
•	Large header – 20 bytes	Small header – 8 bytes.

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* TCP/IP Model

- The TCP/IP (Transmission Control Protocol / Internet Protocol) is the fundamental communication architecture of internet. It defines how data is packaged, addressed, transmitted, routed & received across networks.

- It is protocol-based, meaning that it specifies a set of rules (protocols) that computers & peripherals like routers, switches, network cards, printers, etc. use to communicate reliably.

- It is practical with 4 layers & is the one actually used in real-world meaning.

→ Layers of TCP/IP Model:-

1) Application Layer: Closest to the user. Provides network services to applications. Defines how software applications communicate over a network. Handles high-level protocols for file transfer, email, browsing, etc. Protocols used are HTTP/HTTPS, FTP, SMTP, DNS.

2) Transport Layer: Provides end-to-end communication between devices (host-to-host). Splits application data into segments. Ensures reliability, sequencing & error-checking. TCP (Transmission Control Protocol) & UDP (User Datagram Protocol). Peripherals involved are operating system kernel (TCP/UDP stacks), NIC (Network Interface Cards)

3) Internet Layer: Responsible for logical addressing & routing. It assigns IP addresses & determines best path from source to destination across multiple networks. IP (Internet Protocol) used for addressing & routing. Peripherals involved are routers, Layer 3 switches, firewalls

4) Network Access Layer (or Link Layer): Responsible for physical transmission of data over the network. Converts IP packets into frames & transmits over physical media (cables, wifi, fiber). Handles MAC addresses, error detection at hardware level. Peripherals involved are Network Interface cards (NICs), hubs, switches, modems, access points, physical cables.

* Network & its types

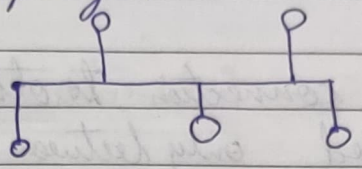
- A network is a collection of two or more interconnected devices (servers, computers, printers, smartphones or peripherals) that communicate & share resources like data, files, internet, applications using transmission media & communication protocols.

→ The internet is a type of world-wide computer network. The internet is a connection of infinite numbers of connected computers that are spread across the world.

→ Network topology is the arrangement of the various components like links, nodes, etc of a computer network.

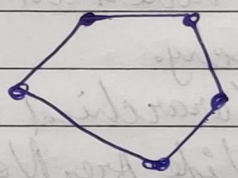
* Types of network topologies :

1) Bus Topology



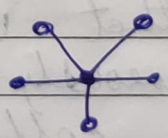
- Every computer & network device is connected to a single cable.
- It transmits data only in one direction.
- Cost effective
- Used in small networks.
- Easy to expand joining two cables together
- It is used in early LAN connection.

2) Ring Topology



- It forms a ring as each computer is connected to another computer, with the last one connected to the first.
- Transmission is unidirectional & sequential way that is bit by bit.
- Transmitting network is not affected by high traffic or by adding more nodes, as only the nodes having tokens can transmit data.
- Cheap to install & expand.

3) Star Topology



- Computers are connected to a single central hub through a cable.
- Fast performance with few nodes & low network traffic.
- Easy to troubleshoot & easy to setup & modify.
- Hub can be upgraded easily.

4) Mesh Topology

- Point-to-point connection to other devices or fully connected.
- Traffic is carried only between two connected devices.
- Robust, costly but not flexible.
- Fault is diagnosed easily.
- More cable resource used in setup.

5) Tree Topology

- It has a root node & all other nodes are connected to it forming a hierarchy.
- Also called Hierarchical topology
- Mostly used in Wide Area Network - WAN.
- Expansion of nodes is possible & easy.
- Easily managed & maintained.

* CSMA/CD (Carrier Sensor Multiple Access with Collision Detection)

- It is a network protocol used in Ethernet to manage how devices share the same communication channel. It ensures that when multiple devices try to send data simultaneously on a shared medium, the system can detect collisions and resolve them.

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Q1)

* **Uses of Computer Networks:-**

- Hardware devices like printers, scanners & external storage can be shared across multiple computers.
- Instead of installing software on every computer, centralized applications can be run from servers.
- Users can easily share files, documents & media across the network.
- Networks enable instant communication through email, chat apps & video conferencing.
- Services like Skype, WhatsApp & Zoom rely on networks to transmit voice & video over data using Voice over IP
- Teams can work on shared documents & projects in real time like Google Docs.
- Large organizations store & manage data in central servers for easy access & security.
- Cloud Computing services like AWS, Google Cloud & Microsoft Azure allow businesses & individuals to use powerful computing resources without owning the hardware.

Q2) Define computer network & list 3 common applications

- A computer network is a system in which two or more computer networks & peripheral devices are connected together using communication channels (wired or wireless) to share resources, exchange data and enable communication.

Three common applications are :-

- Users can share files, printers & storage devices across the network.
- Email, messaging & video conferencing are enabled through computer networks.
- Networks provide access to the internet, cloud computing, e-commerce & multimedia streaming.

Q3) Uses of bridges in computer network.

Ans3) A bridge is a networking device that connects & filters traffic between two or more LAN segments. It works at the Data Link Layer (Layer 2) of the OSI model. Its main role is to forward useful data from one network segment to another while reducing unnecessary traffic.

- Bridges divide a large network into smaller segments which reduces traffic congestion by limiting unnecessary data broadcasts.

- Traffic Filtering: Bridges analyze the MAC addresses of data packets. They only forward data to segment where the destination device exists. This improves efficiency & reduces collisions.
- Extending Networks: Bridges can connect two physically separate LANs, making them act as one large network. Useful when expanding office spaces or linking different floors of a building.
- By reducing collision domains, bridges improve overall network speed & performance.
- Bridges are usually protocol-independent. They simply forward data based on MAC addresses, regardless of higher-level protocols (like TCP/IP).

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24 September 2025 – Wednesday

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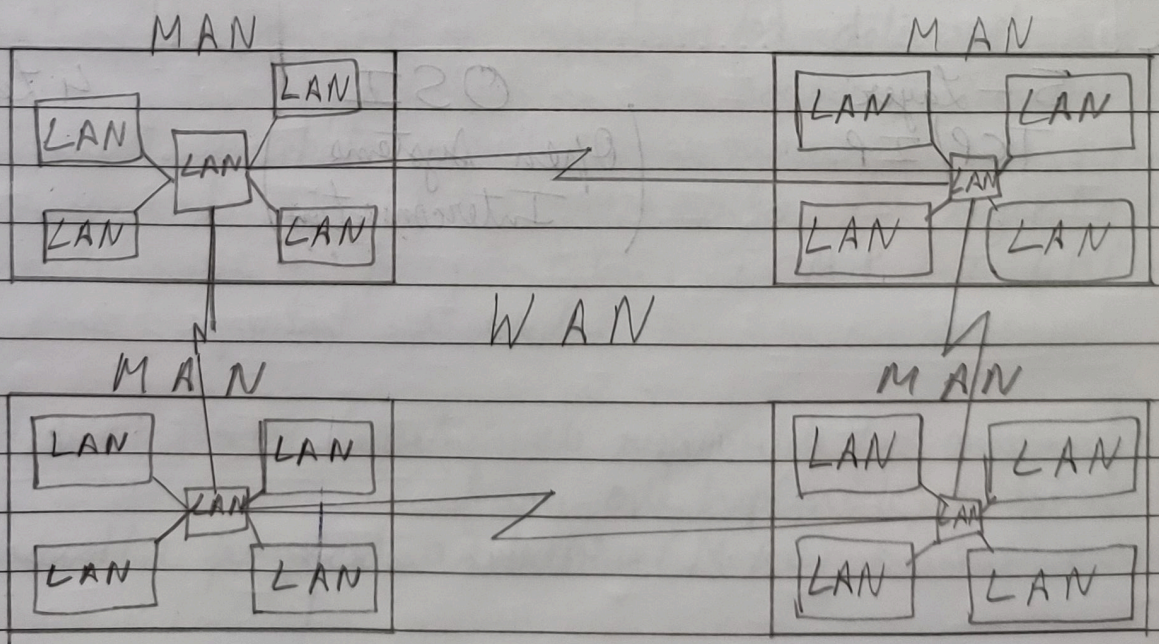
- A computer network is a system in which multiple computers are connected to each other to share information & resources.
- Connections can be wired (cables, switches, routers) or wireless (wifi, Bluetooth). It also includes network devices that help in communication.

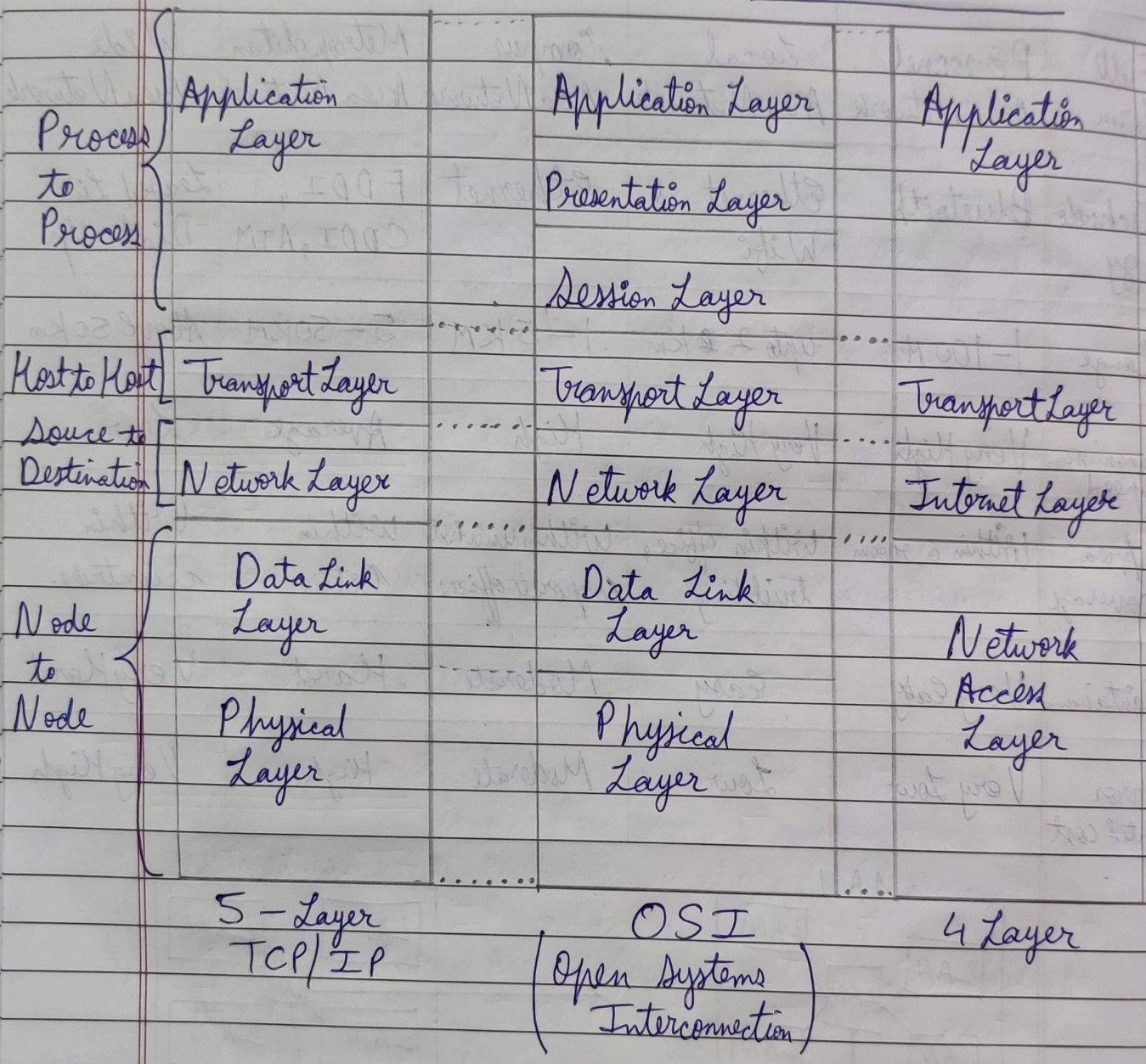
* Uses

1) Better communication i.e. easy exchange of data/ message.
2) File sharing i.e. multiple users can share files & data.
3) Resource sharing i.e. Printers, storage devices & internet can be shared.
4) Internet Access i.e. Connection to the Global Network
5) Flexible Access i.e. access resources from anywhere.
6) Entertainment i.e. streaming, gaming, multimedia
7) It reduces cost of hardware / software by sharing.

	PAN	LAN	CAN	MAN	WAN
Full form	Personnel Area Network	Local Area Network	Campus Area Network	Metropolitan Area Network	Wide Area Network
Technology	Bluetooth	Ethernet & Wifi	Ethernet	FDDI, CDDI, ATM	Leased Line Dial-Up
Range	1-100 M	upto 2 Km	1-5 KM	5-50KM	Above 50 km
Transmission Speed	Very High	Very High	High	Average	Low
Area coverage	Within a room	Within office, building	Within univisty corporate offices	Within Cities	Within countries.
Maintainance	Very Easy	Easy	Moderate	Hard	Very Hard
Error Rate & cost	Very Low	Low	Moderate	High	Very High



* TCP/IP Protocol Suite or Internet Protocol



* 4 Layer Model

1) **Application Layer**: It provides network services directly to user applications (web browsers). This is allowing access to network resource. It provides services like network virtual terminal, file transfer, access & managment, mail services, etc.

2) **Transport Layer**: It is responsible for the delivery of message from one process to another. This layer ensures that the whole message arrives intact & in order. It provides services like multiplexing/demultiplexing, reliability, flow control & segmentation/reassembly. Two core protocols are :-

   1) TCP – Transmission Control Protocol

   2) UDP – User Datagram Protocol.

3) **Internet Layer**: It is responsible for delivery of individual packets from the source to ~~host~~ destination. It routes packets across one or more networks. Primary protocol is IP, IP is responsible for logical addressing, packetization & fragmentation. ICMP & IGMP are other important protocols.

4) **Network Access Layer**: It is responsible for moving frames from one hop(node) to the next. It handles physical addressing (MAC), framing, error detection.

* OSI Model (7 layers)

- Developed by International Standards Organization (ISO)

1) **Physical Layer:** It is responsible for movements of individual bits from one hop (node) to the next. It carries the bit stream over a physical media. It is concerned with representation of bits, data rate, synchronization of bits, physical topology.

2) **Data Link Layer:** It is responsible for moving frames from one hop (node) to the next. It provides error detection, correction & framing. It organizes raw bits into frames carried from physical layer. It is responsible for framing, physical addressing (MAC addresses), flow control, error control, and access control.

3) **Network Layer:** It is responsible for delivery of individual packets from the source to destination. It is responsible for logical addressing eg 192.68.1.1 (IP address) and routing of packets between devices on different networks. Examples are IPv4, IPv6, ICMP (ping), IGMP, routers.

4) **Transport Layer:** It delivers the message from one process to another. It provides end-to-end communication between applications. It is responsible for segmentation & reassembly of data. It ensures that the whole message arrives intact & in order. Examples are TCP (Transmission Control Protocol) & UDP (User Datagram Protocol).

5) **Session Layer:** It is responsible for dialog control & synchronization. This layer acts as a network dialog controller — establishes, maintains, synchronizes the interaction among computers. It is the manager of sessions between two systems.

6) **Presentation Layer:** It is responsible for data translation, compression, encryption. This layer is concerned with the syntax which refers to order in which data is presented & semantics helps in interpreting a particular pattern.

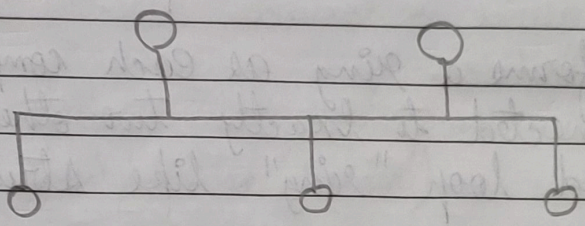
7) **Application Layer:** It is responsible to provide network services to the user. It provides services like file transfer, access & management, mail services, directory services. It defines high level protocols like HTTP/HTTPS, FTP, SMTP, DNS

	OSI Model	TCP/IP Model
1)	It has 7 layers	It has 5 layers
2)	It provides layer functioning. Each layer has specific, well defined ~~fun~~ functions.	It is based on protocols & protocols are not flexible with other layers.
3)	The Transport layer gaurantees delivery of packets.	The Transport layer does not gaurantee delivery of packets.
4)	It follows horizontal way.	It follows vertical way.
5)	More prescriptive: each layer has a narrowly defined role.	Less rigid — some functions are spanning layers.
6)	It provides both connection oriented & connectionless service in	It provides connectionless service.
7)	Protocols are hidden OSI layers & are easily replaced	Protocols are explicit & cannot be replaced easily
8)	It defines services, interfaces & protocols very clearly & makes clear distinction between them.	It does not clearly seperates its services, interfaces & protocols.

* Topologies

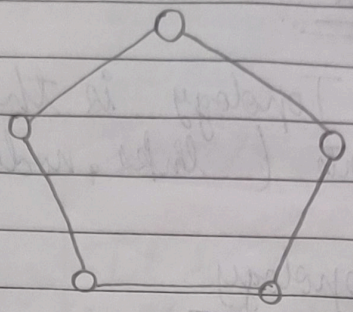
Network Topology is the arrangement of the various components (links, nodes, etc) of a computer network.

1) Bus Topology



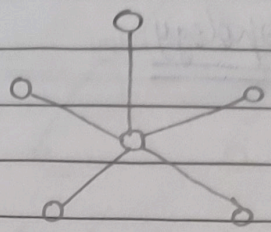
- Every computer & network device is connected to a single cable.
- It transmits data only in one direction.
- Cost effective
- Used in small networks
- Easy to expand, by joining two cables together.
- It is used in early LAN connection.

2) Ring Topology



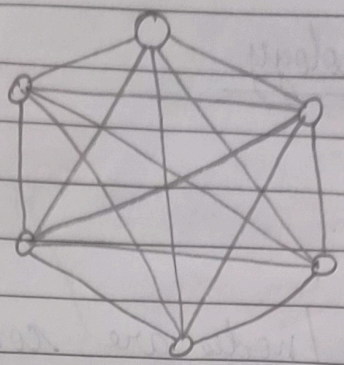
- It forms a ring as each computer/device is connected to exactly two other devices, forming a closed loop "ring" like structure.
- Data travels in one direction or both directions, passing through each device until it reaches its destination.
- Simple & cheap to install & expand
- Data moves in circular path.
- Performs better than bus topology under heavy network load.
- No central hub/switch is required unlike star topology.
- Failure of a single node/cable can break the entire network.
- Difficult to troubleshoot & reconfigure.
- Slower than star topology in large networks.

3) Star Topology



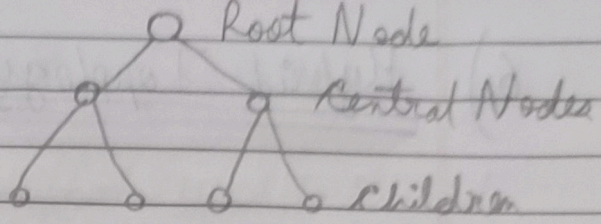
- All devices / nodes are connected to a central device through a cable.
- Communication between two nodes passes through the central device (hub / switch / router).
- Each node has a point - to - point connection to the central device.
- Failure of one node does not affect the rest of network.
- Centralized managment makes troubleshooting easy.
- Provides better performance than bus or ring since nodes don't share a single cable.
- If the central hub / switch fails, the whole network stops.
- Requires more cable than bus or ring topology.
- Cost is higher due to central networking devices.
- Example: Most modern ethernet LANs use star topology with switches. Common in office networks.

4) Mesh Topology



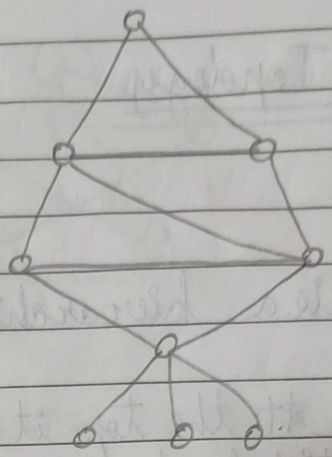
- Every device is connected directly to every other device in the network.
- It has multiple paths for data travel, ensuring high redundancy & reliability.
- It provides maximum redundancy but is expensive.
- Connection is carried only between two connected devices.
- Fault is diagnosed easily.
- More cable resource used in setup.
- Complex setup & maintenance ; harder to manage.
- Not scalable for very large networks.
- Robust & fast communication.
- Highly reliable : even if one link fails, data can take another path.

5) Tree Topology



- It is a hierarchial network topology.
- ~~It~~ At the top it has root node, below it are multiple levels of secondary node connected in a ~~star~~ tree like manner.
- It has root node connected to one or more central nodes.
- Central nodes act as parents & devices connected to them act as children.
- Expansion happens level by level, like branches of a tree.
- Easy to add more nodes by extending branches.
- Each level can be managed separately.
- If one branch fails, only that branch is affected.
- If Root node fails, entire network fails.
- Complex installations & maintenance as the network grows.
- It is a combination of star & bus.

6) Hybrid Topology



- It is combination of 2 or more different network topologies like star, bus, mesh, tree, ring to form a single, efficient network.
- It is designed to take the advantages of each topology & minimize their disadvantages, making it the most optimized network for large & complex ones.
- It provides flexibility & scalability.
- Design depends on the organization's needs.
- Highly reliable; a failure in one part doesn't necessarily break the whole network.
- Complex design & installation
- Hight cost (cabling + device)
- Maintenance is difficult due to multiple types of connections.

* Carrier Sense Multiple Access (CSMA)

It is a network access method used in shared communication channels like Ethernet or Wifi where multiple devices share the same medium.

Before sending data, a device "listens" (carrier sense) to check if the channel is free.

- If the medium is idle, it transmits.
- If the medium is busy, it waits until it becomes free.

This helps reduce collisions when multiple devices try to send data at the same time.

→ Carrier Sense: Each device listens to the medium before transmitting.

→ Multiple Access: Many devices share the same communication channel.

→ Collision Possibility: Even with listening, two devices may still transmit almost simultaneously in which ~~result~~ collision may occur.

- It reduces collisions compared to pure random access like ALOHA.
- It is simple & decentralized. No ~~extend~~ central controller needed.

* Carrier Sense Multiple Access / Collision Detection (CSMA/CD)

CSMA/CD is a media access control method used mainly in wired ethernet networks to manage how multiple devices share the same communication channel.

It combines two ideas:

1) CSMA (Carrier Sense Multiple Access): each device listens to the channel before transmitting.

2) CD (Collision Detection): If two devices still transmit at the same time and a collision occurs, they detect it, stop and retransmit later.

* Working

- Device checks if the channel is idle.
- If free, it starts transmitting.
- While transmitting, it listens for changes to detect if another device is also transmitting.
- If collision detected, devices send a jam signal to notify others.
- Each device waits for a random backoff time
- After waiting, devices sense the medium again and retransmit.

* It prevents data corruption by detecting collisions early.

25 - September - 2025 — [redacted]

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* Carrier Sense Multiple Access with Collision Avoidance

CSMA/CA is a media access control method mainly used in wireless networks (wifi) to reduce the chance of collisions when multiple devices share the same channel.

Unlike CSMA/CD, wireless devices cannot detect collisions while transmitting. So instead of detecting, CSMA/CD tries to avoid collisions before they happen.

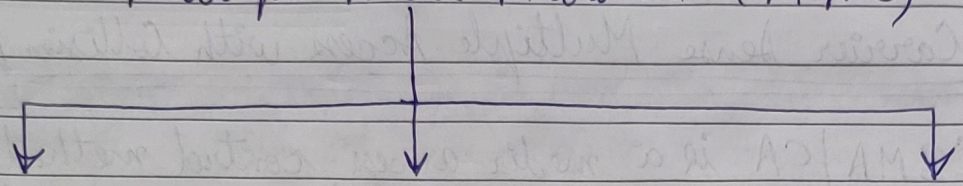
* Working

- Device listens to the channel to check if its free.
- If idle, device waits for a short period, then sends data.
- If busy, device waits for a random backoff time before checking again.
- Reciever sends an ACK frame if data is recieved correctly, ensuring reliability.

* Advantages

- It reduces chances of collisions in wireless networks.
- Works well with many devices in the same wifi area.
- Supports reliability with ACK mechanism.

Multiple Access Protocols (MAC)



Random Access Protocols	Control Access	Channelization Protocols
• Aloha	• Polling	• FDMA
• CSMA	• Token passing	• TDMA
• CSMA/CA		• CDMA
• CSMA/CD		

* Pure Aloha

- It is a simple, random access protocol used in computer networks where multiple devices share a common communication channel.
- In pure ALOHA, a device can transmit data at any time without checking if channel is free.

* Working

1) A device sends a frame whenever it has data.
2) After sending, it waits of an acknowledgement (ACK) from the reciever.
3) If ACK is received i.e. transmission succesful.
4) If no ACK i.e. collision occured & device waits

a random backoff time & retransmits.

* Since devices don't sense the medium before sending, collisions are frequent.

- The throughput of the Pure ALOHA is maximized when the frames are of uniform length.
- The formula to calculate throughput of the Pure ALOHA is

$$S = G * e^{-2G}$$

Here, G = number of stations transmitting data
e = constant

Throughput is maximum when $G = 1/2$ which is 18% of the total transmitted data frames.

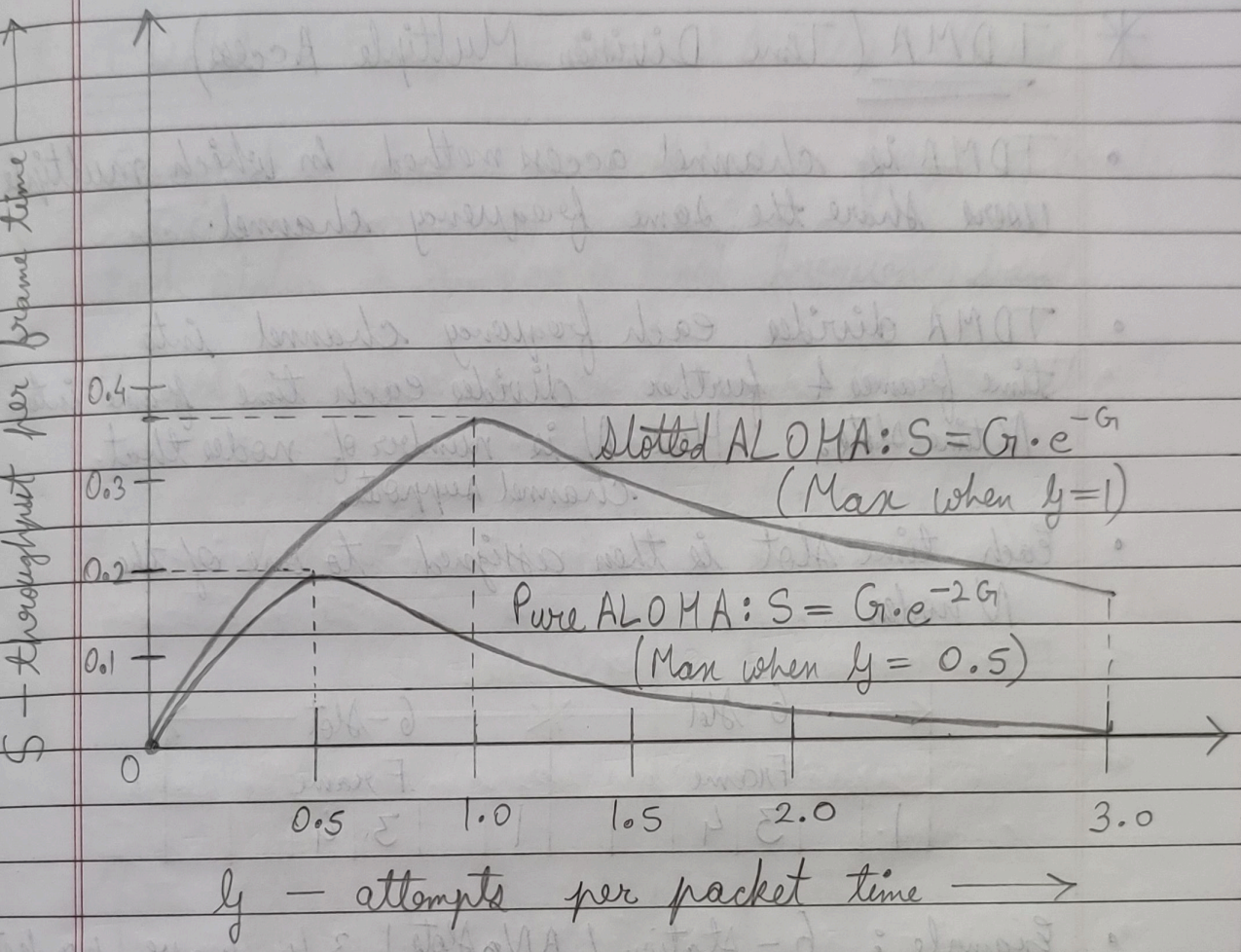
- Works well with low traffic networks
- Useful in satellites & wireless systems with long propagation delays.
- High collisions probability under heavy load.
- ~~No~~ Not efficient for large scale networks.

* Slotted ALOHA

- It was invented to improve the efficiency of pure ALOHA as chances of collisions in pure ALOHA are very high.
- The time of the shared channel is divided into discrete intervals called slots.
- The stations can send a frame only at the beginning of the slot & only one frame is sent in each slot.
- If any station is not able to place the frame onto the channel at the beginning of the slot then the station has to wait until the beginning of the next time slot.
- The formula to calculate the throughput of the Slotted ALOHA is

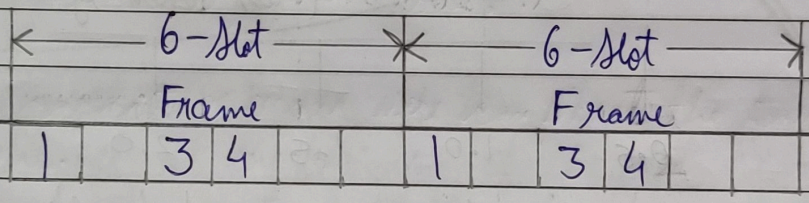
$$S = G * e^{-G}$$

- The throughput is maximum when $G = 1$ which is 37% of total transmitted data frames.
- 37% of the time slot is empty
- 37% successes
- 26% collisions


S – throughput per frame time
0.4
0.3
0.2
0.1
0
Slotted ALOHA: $S = G \cdot e^{-G}$
(Max when $G = 1$)
Pure ALOHA: $S = G \cdot e^{-2G}$
(Max when $G = 0.5$)
0.5
1.0
1.5
2.0
3.0
G – attempts per packet time

* TDMA (Time Division Multiple Access)

- TDMA is channel access method in which multiple users share the same frequency channel.
- TDMA divides each frequency channel into time frames & further divides each time frame into N time slots. Here N is number of nodes that channel supports.
- Each time slot is then assigned to one of the N nodes.



- Example: 6-station LAN slots 1, 3, 4 have packet slots 2, 5, 6 are idle.

* Advantages

- No collision since slots are pre-allocated.
- Efficient use of bandwidth in high-load systems.
- Simple scheduling for multiple users.

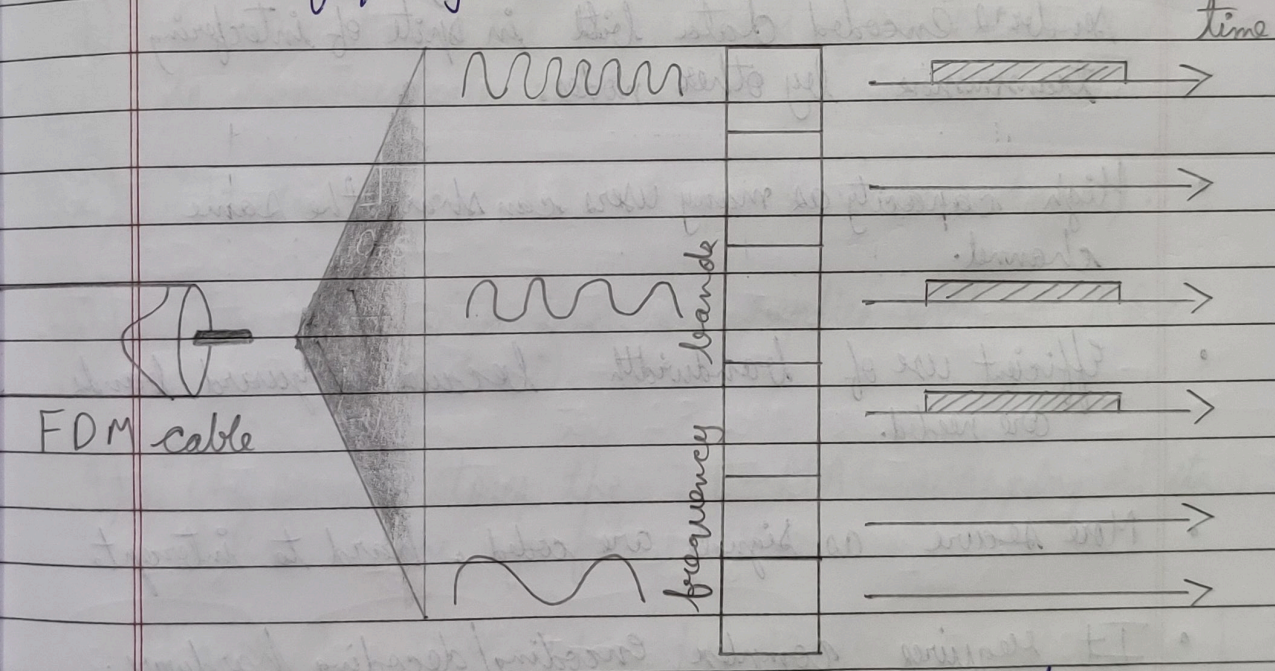
* Disadvantages

- Requires time synchronization between devices.
- If a user has no data to send, their slot is wasted.
- Higher complexity compared to ALOHA.

27 September 2025 — Saturday

* FDMA — Frequency Division Multiple Access

- Channel spectrum is divided into frequency bands.
- Each station is assigned a fixed frequency band.
- Unused transmission time in frequency bands go idle.
- Example: 6-station LAN. 1, 3, 4 have packet, frequency bands 2, 5, 6 are idle.



- All users can transmit simultaneously, but on different frequency channel & so no interference occurs.
- No collisions as each user has its own channel.
- Works well for continuous, steady traffic (like voice)
- Examples are Radio & TV broadcasting, 1G analog mobile systems, Satellite communication.

* CDMA — Code Division Multiple Access

- It assigns a unique code to each node, each node then uses its unique code to encode the data bits it sends.
- If the codes are chosen carefully, CDMA networks can transmit simultaneously from different nodes.
- Their respective receivers correctly receive a sender's encoded data bits in spite of interfering transmissions by other nodes.
- High capacity as many users can share the same channel.
- Efficient use of bandwidth because no guard bands are needed.
- More secure as signals are coded, hard to intercept.
- It requires complex encoding/decoding hardware.

* ARP — Address Resolution Protocol

ARP is a network protocol used to find the MAC address i.e. physical address of a device from its IP address in a Local Area Network (LAN).

ARP maps an IP address to a MAC address.

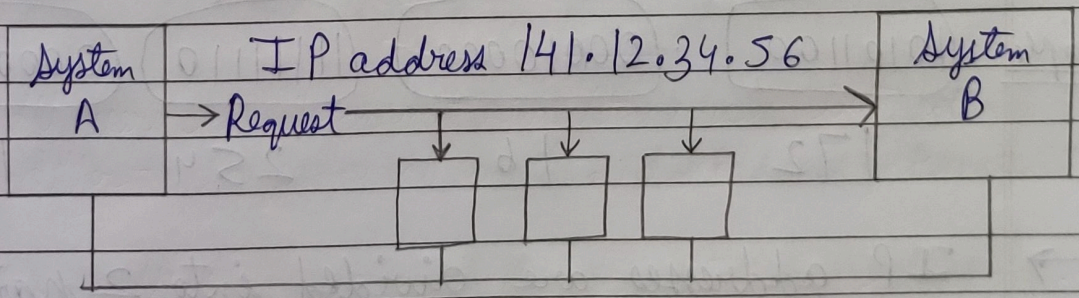
* Working

1) A device knows the IP address of another device it wants to communicate with.

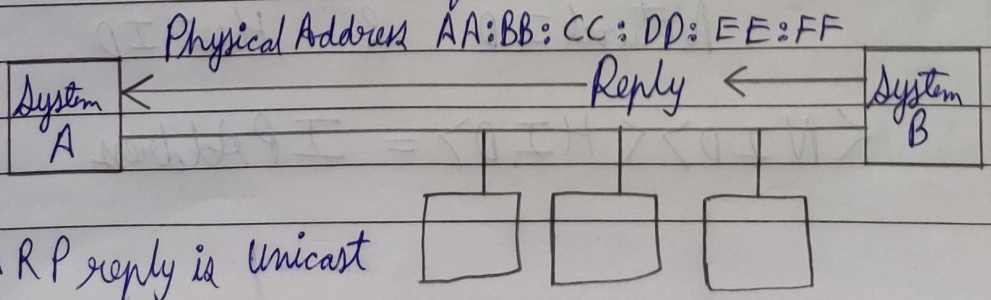
2) It sends an ARP Request (broadcast).

3) The device with that IP replies with an ARP Reply (unicast).

4) The sender stores the IP-MAC mapping in its ARP cache for future use.



ARP request is broadcast.



ARP reply is unicast

* IP Address (Internet Protocol Address)

An IP address is a unique numerical identifier assigned to every device connected to a network like the internet or a LAN.

It helps in identifying a device & locating it on the network.

* Functions

- Identification of a device on a network.
- Shows where the device is located in the network.
- Routing the data packets find the correct destination.

→ IPv4 — IP addresses are 32 bit numbers which are divided into 4 octets. Each octet represents 8 bit binary number.

Example

8 bit	8 bit	8 bit	8 bit
10101100	00010000	11111110	00000001
172	16	254	1

→ IP addresses are divided into 2 parts:

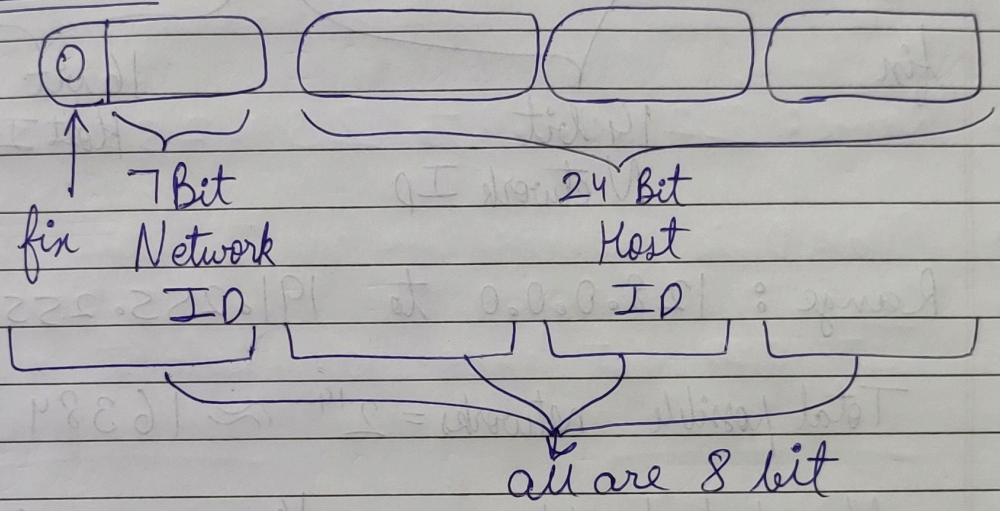
Network ID & Host ID

<NID><HID> = IP Address

* Classes of IPv4 addresses

IPv4 addresses are divided into 5 classes A to E
(32 bit — Dotted decimal format)

1) Class A

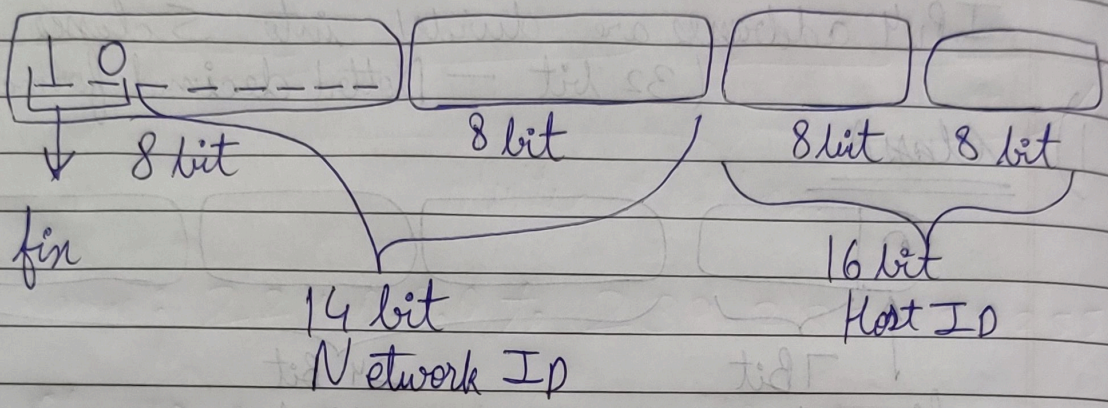


- Range : 1.0.0.0 to 126.255.255.255
(Useful)
- Only 126 addresses are used for network address.
- Host IPs per network = $2^{24} - 2$
- 0.0.0.0 — Special IP address — Network Number
127.255.255.255 — Special IP address — Broadcast IP.

* Default Subnet Mask = 255.0.0.0 (/8)

CIDR Notation

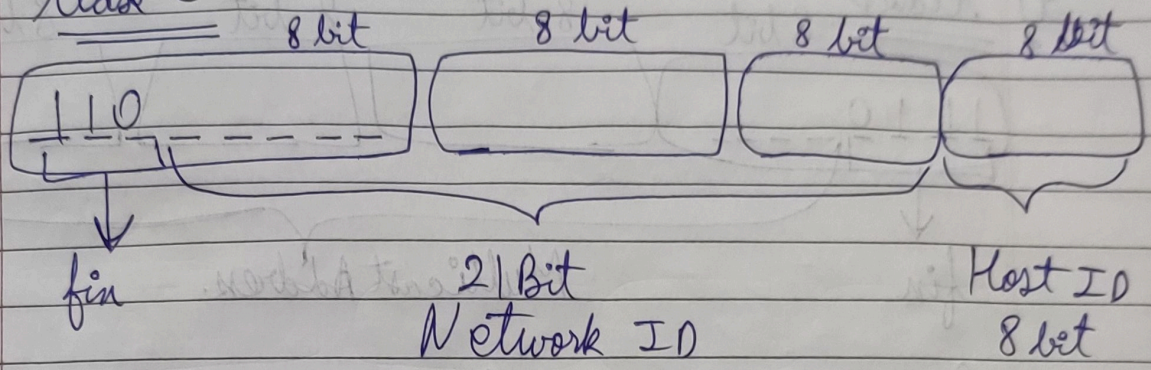
2) Class B



- Range : 128.0.0.0 to 191.255.255.255
- Total possible networks = $2^{14} \approx 16384$
- No of host addresses = $2^{16} - 2 \approx 65534$

☆ Default Subnet Mask = 255.255.0.0 (/16)

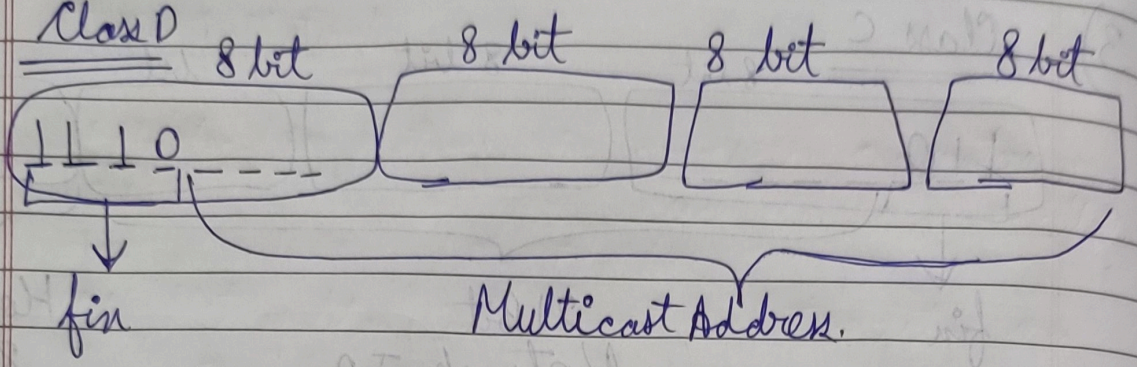
3) Class C



- Range: 192.0.0.0 to 223.255.255.255
- No. of total networks = $2^{21} \approx 2M$
- Hosts per network = $2^8 - 2 = 256 - 2 = \boxed{254}$

☆ Default Subnet Mask = 255.255.255.0 (/24)

4) Class D



- Range: 224.0.0.0 to 239.255.255.255
- Class D is reserved for multicasting
- It does not have any subnet mask.

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6 October 2025 — Monday

* Uses of Computer Network

- Computer in a network can share hardware & software resources, reducing cost & improving efficiency.
- Users can easily share & access data, files & information across the network
- Ensures electronic communication through emails, chat, video conferencing

* A computer network is a system of computers interconnected to each other through wires or wireless to share information & resources, exchange data.

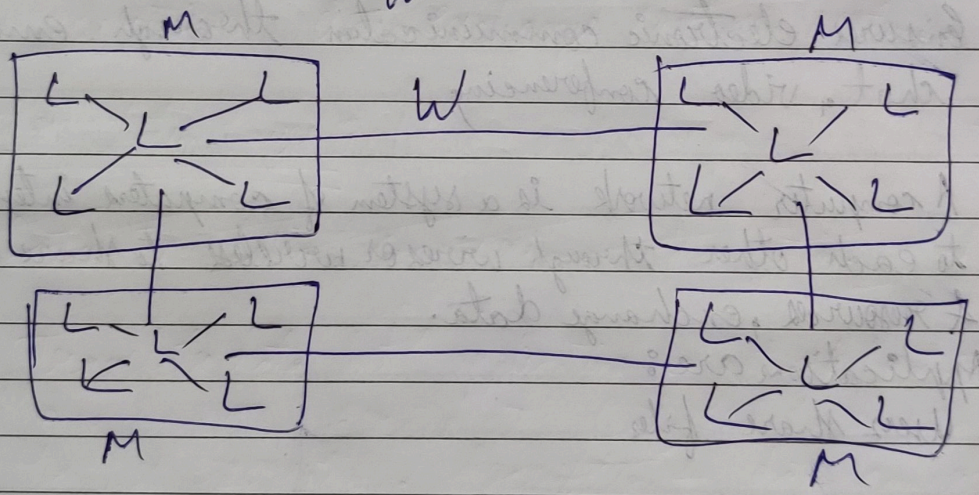
Applications are :-

- Users share files
- Email, message, video conferen
- E-commerce, multimedia stream
- Flexible access from anywhere
- Cloud computing

* Types of Network

- LAN (Local Area Network) interconnects computers within a limited area such as a school or college
- MAN (Metropolitan) interconnects in metropolitan area like city.
- WAN (Wide) it exists over a large-scale geographical area.

WAN connects different LANs & MANs



* OSI Model — Open Systems Interconnection.

7	Application
6	Presentation
5	Session
4	Transport
3	Network
2	Data Link
1	Physical

1) Physical Layer:

- Transmits raw bits over a physical medium
- It is responsible for representation of bits, data rate, synchronization of bits.

2) Data Link Layer:

- Responsible for ~~the~~ moving frames from one node to next.
- Handles flow control, error control, access control.

3) Network Layer:

- Determines best path for data routing.
- Provides logical addressing — IP addresses.
- Responsible for delivery of packets.

4) Transport Layer:

- Delivers message from one process to another.
- Ensures reliable data transfer.
- TCP & UDP are examples

5) Session Layer:

- Establishes, manages & terminates sessions.
- Synchronizes data exchange between systems.

6) Presentation Layer:

- ~~Data translation, compression & encryption.~~
- Translates data bet" applications & network formats.
- Handles encryption, decryption & data compression.

7) Application Layer:

- Provides user interfaces & network services
- Supports protocols like HTTP, FTP, SMTP.

* TCP/IP

4 Application Layer

3 Transport Layer

2 Internet Layer

1 Network Access Layer

1) Network Access Layer

- Deals with physical transmission of data
- Defines how data is sent over physical networks
- Handles framing, error detection & physical addressing (MAC)

2) Internet Layer

- Responsible for logical addressing & routing.
- Determine best path for data packets.
- Protocols used are Internet Protocol, ICMP, ARP.

3) Transport Layer

- Provides end-to-end communication bet^n devices.
- Handles segmentation, flow control & error correction.
- Ensures ~~exist~~ reliable data delivery & fast, connectionless servicd.

4) Application Layer

- Provides services directly to user/applications
- Protocols: HTTP, FTP, DNS, SMTP
- Enables communication bt" software applications over the network.

* MAC address — Media Access Control

- It is a unique hardware identifier assigned to every network interface card (NIC) by the manufacturer.
- It uniquely identifies a device on a local network.
- Used in data transmission within same local area network (LAN).
- MAC Address operates at Data Link Layer (Layer 2) of the OSI Model